Exercise:

Inertial Measurement Unit „I2C“

Inertial Measurement Unit (IMU):

- Inertial Measurement Unit (IMU), is an electronic device that measures and reports on an object's velocity, orientation, and gravitational forces, using a combination of accelerometers, gyroscopes, and magnetometers.

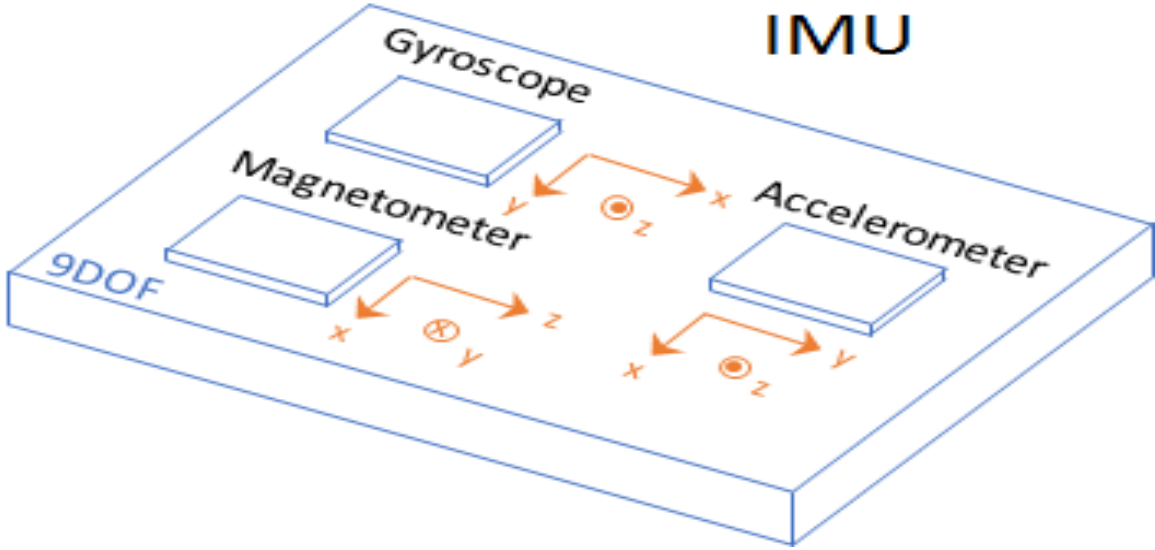
- The LSM9DS1 is an inertial measurement unit (IMU) that packs an 3-axis Gyroscope with 3-axis Accelerometer and 3-axis Magnetometer:

* SPI / I2C serial interfaces.

* 16-bit data output per axis.

* Sensitivity range (configurable):
- Gyro: ±245, ±500, or ±2000 dps angular rate full scale.
- Accelerometer: ±2, ±4, ±6, ±8 or ±16 g linear acceleration full scale.
- Magnetometer: ±4, ±8, or ±12 gauss magnetic full scale.

* Embedded temperature sensor.

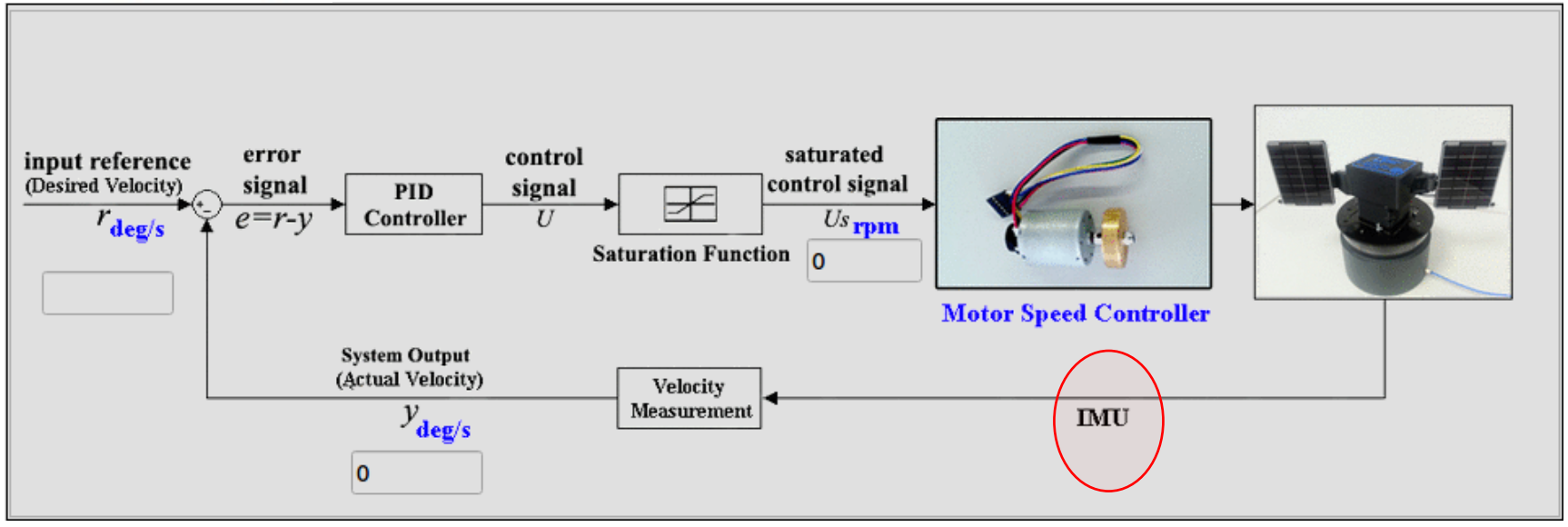


- Orientation Angles:
- Three attitude angles are referenced to the local horizontal plane which is perpendicular to earth's gravity:
* Heading is defined as the angle between the X axis and the magnetic north on the horizontal plane.
* Pitch is defined as the angle between the X axis and the horizontal plane.
* Roll is defined as the angle between the Y axis and the horizontal plane.

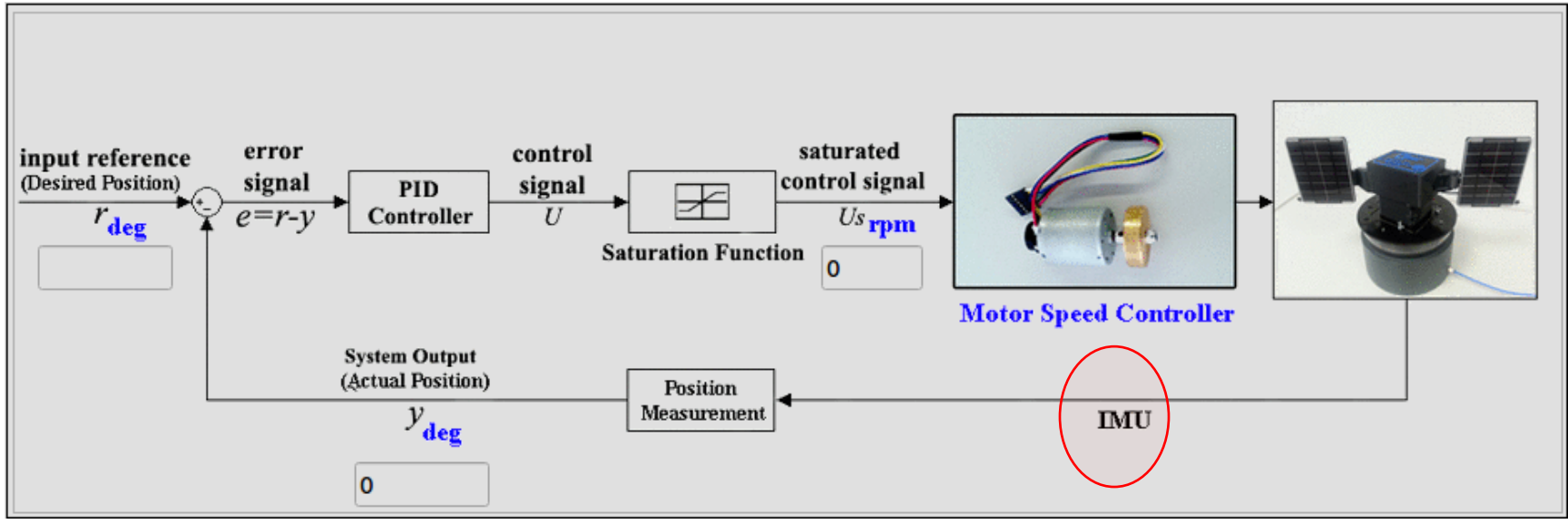
Inertial Measurement Unit (IMU):

Why we need it here??

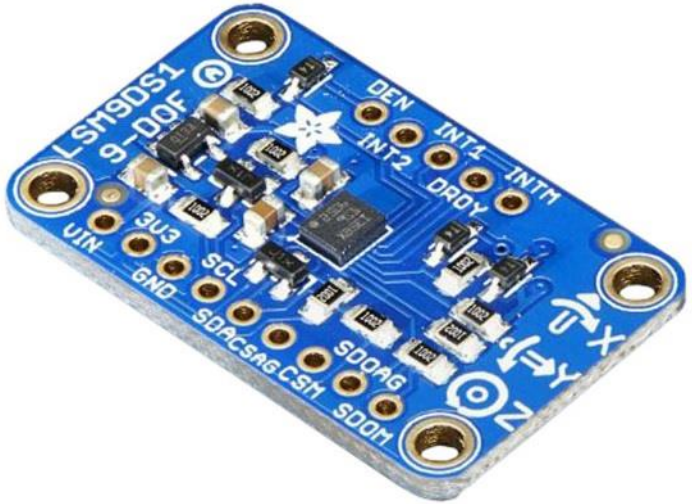
Velocity Controller



Position Controller



Inertial Measurement Unit (IMU):



Inertial Measurement Unit (IMU):

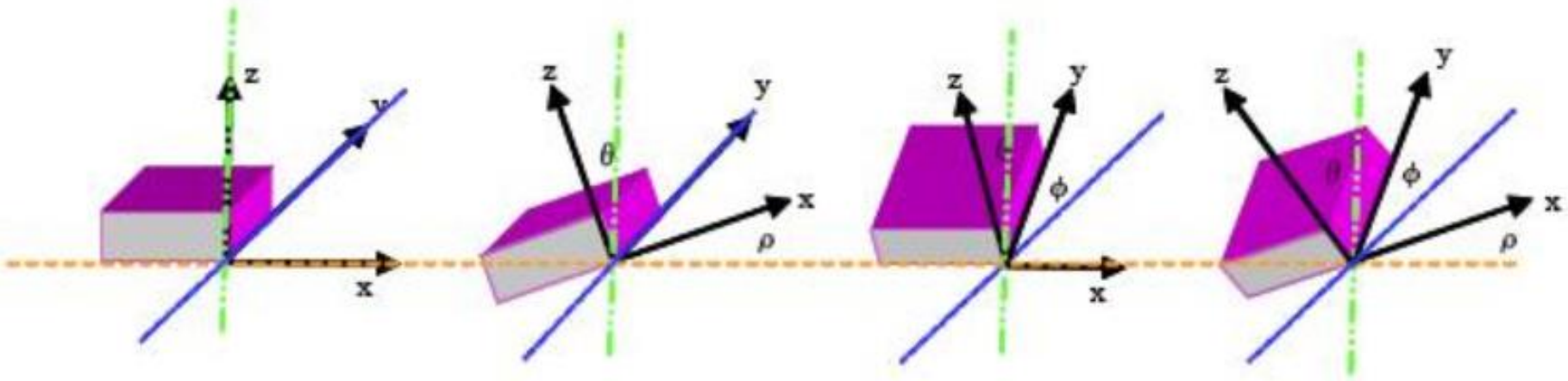
• Accelerometer Sensor:

• Operation:

- Accelerometers are sensors or transducers that measure linear acceleration.

- Accelerometers generally measure linear acceleration forces applied to a body by being mounted directly onto a surface of the accelerated body and also measure the tilt angles of the body: pitch, and roll.

- An accelerometer measures force exerted by a body as a result of a change in the velocity of the body. A moving body possesses an inertia which tends to resist change in velocity. It is this resistance to change in velocity that is the source of the force exerted by the moving body. This force is directly proportional to the acceleration component in the direction of movement when the moving body is accelerated.

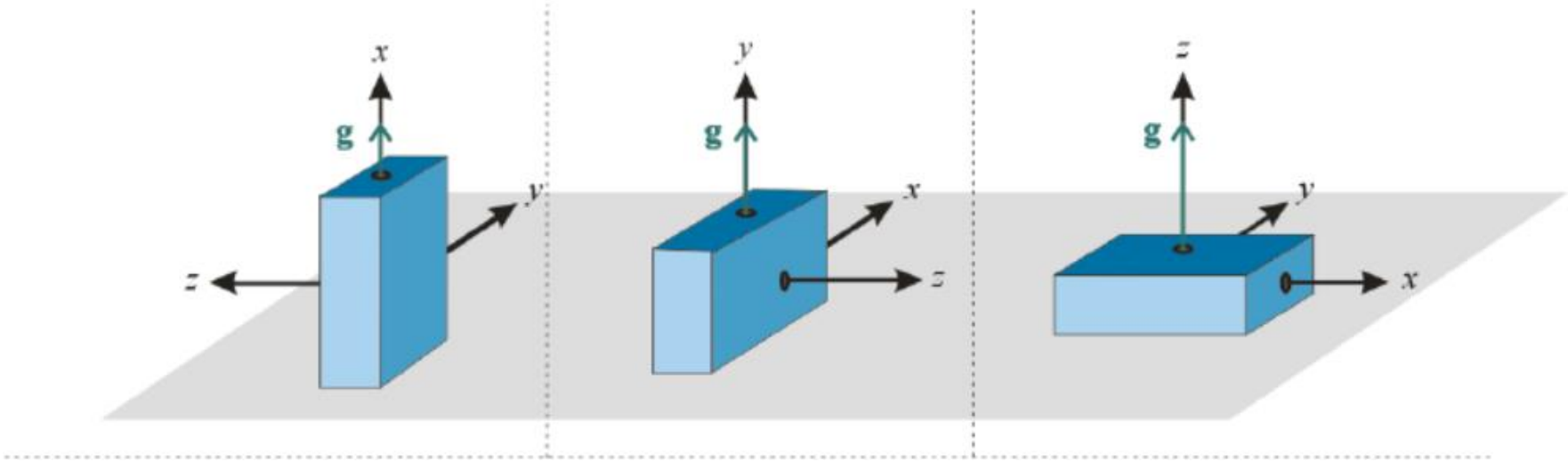


Inertial Measurement Unit (IMU):

• Accelerometer Sensor:

• Calibration:

A simple calibration process for a 3-Axis Accelerometer sensor can be performed by taking N samples for each axis at three positions in standstill condition: P1->[0g X, 0g Y, 1g Z], P2->[0g X, 1g Y, 0g Z] and P3->[1g X, 0g Y, 0g Z] and then average them to get two offset values correspond to 0g measurement for each axis. The two offset values correspond to 0g measurement of each axis can then be averaged to give the final offset values for each axis.



Inertial Measurement Unit (IMU):

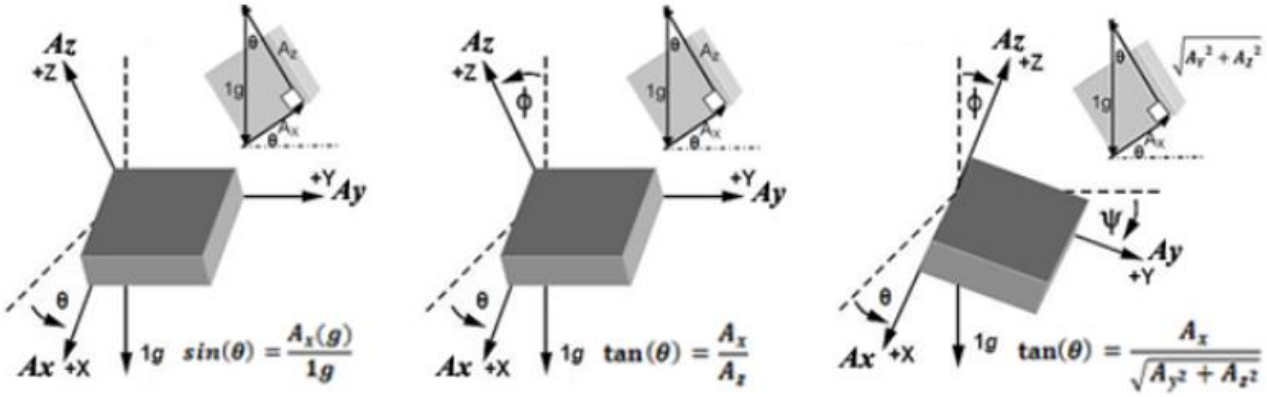
• Accelerometer Sensor:

• Measurement:

There are two types of measurements can be preform using an accelerometer sensor:

* Dynamic Acceleration: In this type of measurement, the sensor measures linear acceleration of a moving body. As The sensor move along an axis, the value corresponds to that axis changes. By knowing the initial offset value of the sensor (standstill value) and by measuring the current value of the sensor continuously, we can know if the sensor is on the move and also we can calculate the acceleration of that movement in g using the sensitivity value provided by the datasheet.

* Static Acceleration: In this type of measurement, the sensor measure the magnitude of earth's gravity field mounted on it, which is very useful to get orientation information of an object. As the sensor change its orientation along an axis, the sensor value corresponds to that axis changes. By knowing the initial offset value of the sensor (standstill value) and by measuring the current value of the sensor continuously, we can know if the sensor has change its orientation along that axis and we can also calculate the angel of that movement using one, two or three axis measurement method:

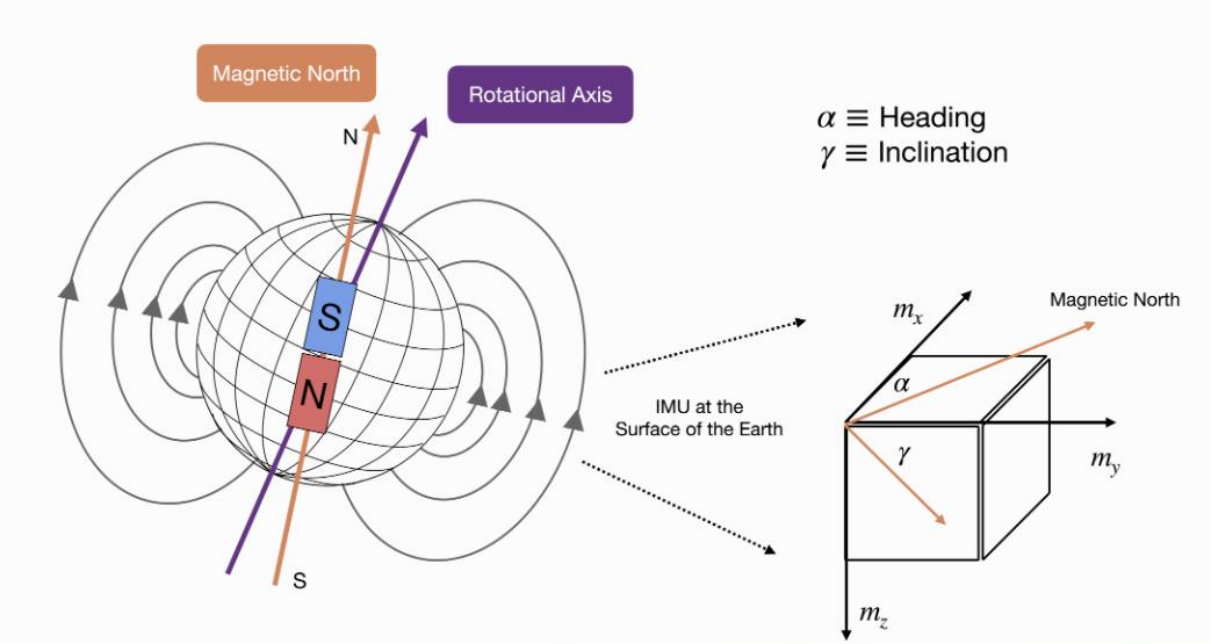


Inertial Measurement Unit (IMU):

• Magnetometer Sensor:

- **Operation:**

  - Magnetometers are sensors that measure magnetic field *B*, which is also known as the magnetic flux density or the magnetic induction.
  - Magnetometers are useful in measuring Earth's magnetic field components which can be used to calculate the magnetic north of the earth (magnetic compass).



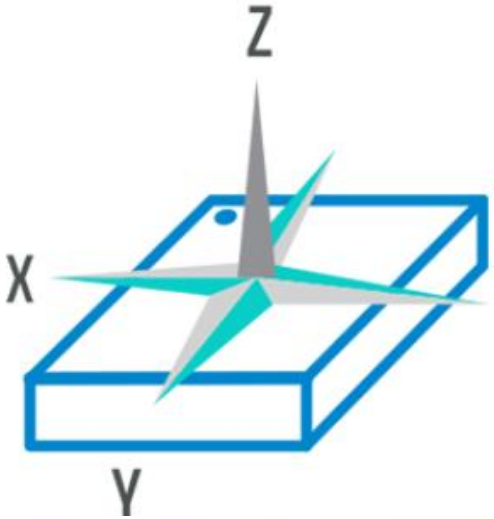
Inertial Measurement Unit (IMU):

• Magnetometer Sensor:

- **Calibration:**

A simple Hard-Iron calibration and normalization process for a 3-Axis Magnetometer sensor can be performed by measuring maximum and minimum values for each axis while rotating the sensor around the axis. These max and min values can be used to shift and scale the axis values as follows:

$$Mx_s = \frac{Mx - Mx_{min}}{Mx_{max} - Mx_{min}} * 2 - 1 \qquad My_s = \frac{My - My_{min}}{My_{max} - My_{min}} * 2 - 1 \qquad Mz_s = \frac{Mz - Mz_{min}}{Mz_{max} - Mz_{min}} * 2 - 1$$



Inertial Measurement Unit (IMU):

• Magnetometer Sensor:

• Measurement:

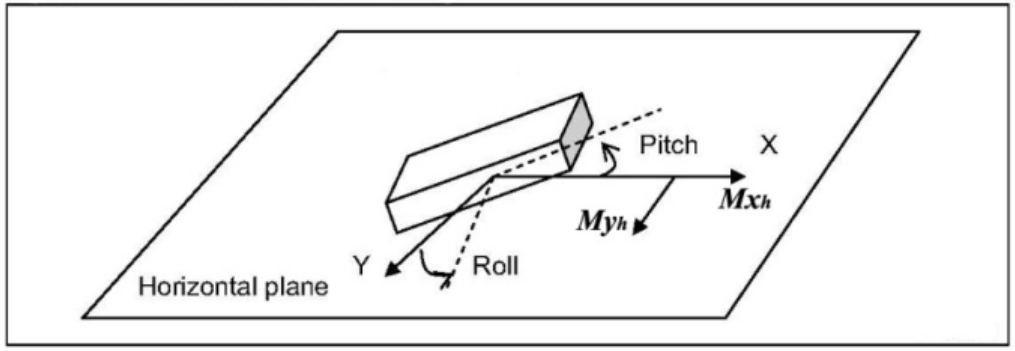
There are two types of measurements can be preform using a magnetometer sensor:

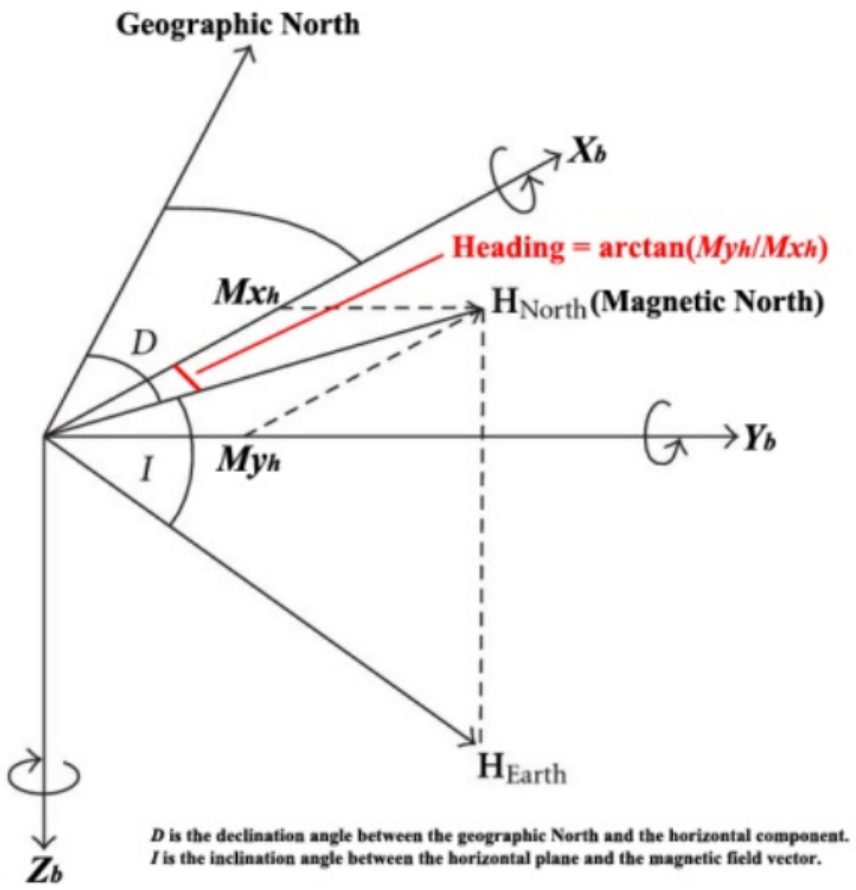
* Magnetic Flux Density: the sensor can measure the magnetic flux density along each axis in Gauss using the sensitivity value provided by the datasheet.

* Compass Angle: the orientation angle of the magnetic north for an object can be calculated using the horizontal components of the magnetometer.

* Tilt Compensation: if the sensor is tilted, the magnetic sensor measurements need to be compensated to obtain the horizontal components Mx_h and My_h using following equations:

$$Mx_h = Mx_s.\cos(pitch) + Mz_s.\sin(pitch)$$
$$My_h = Mx_s.\sin(roll).\sin(pitch) + My_s.\cos(roll) - Mz_s.\sin(roll).\cos(pitch)$$



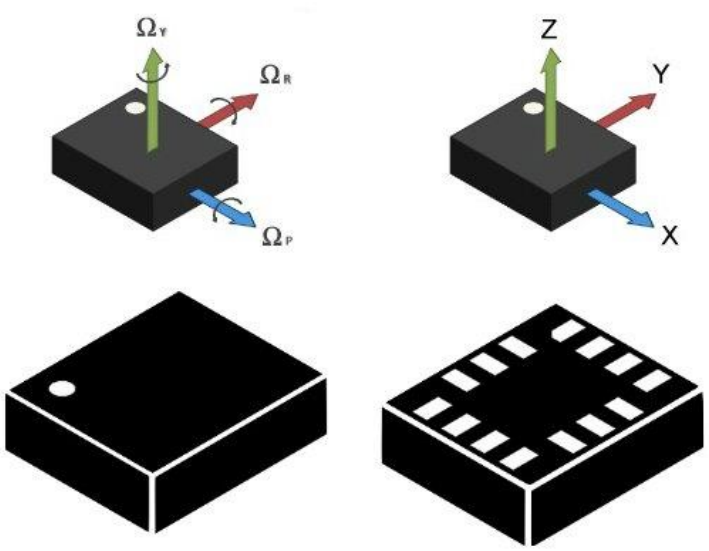


Inertial Measurement Unit (IMU):

● Gyroscope Sensor:

● **Operation:**

- Gyroscopes are sensors or transducers that measure angular velocity.
- Gyroscopes are useful in detecting rotation motion in objects and also in measuring orientation angles of the object: pitch, roll and yaw(heading).
- Microelectromechanical System (MEMS) Gyroscop uses the phenomenon of Coriolis force, which is generated when a rotational angular velocity is applied to the oscillating body where the vibrating object tends to continue vibrating in the same plane as its support rotates.

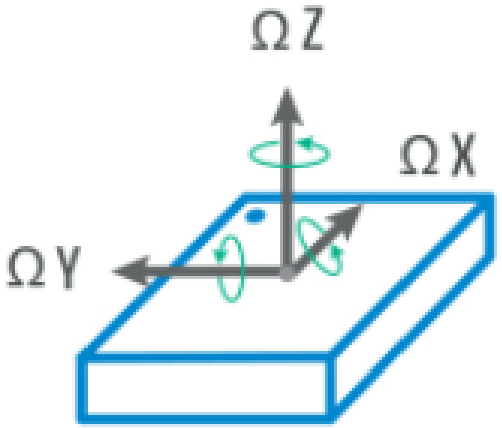


Inertial Measurement Unit (IMU):

● Gyroscope Sensor:

- **Calibration:**

A simple calibration process for a 3-Axis Gyroscope sensor can be performed by taking N samples for each axis in standstill condition and then average them to get the offset value correspond to 0 deg/sec measurement for each axis.



Inertial Measurement Unit (IMU):

● Gyroscope Sensor:

● Measurement:

There are two types of measurements can be preform using a gyroscope sensor:

* Angular velocity: the sensor measures the angular velocity of a rotating body. As The sensor rotate around an axis, the value corresponds to that axis changes. By knowing the initial offset value of the sensor (standstill value) and by measuring the current value of the sensor continuously, we can know if the sensor is rotating and also we can calculate the angular velocity of that movement in deg/sec using the sensitivity value provided by the datasheet.

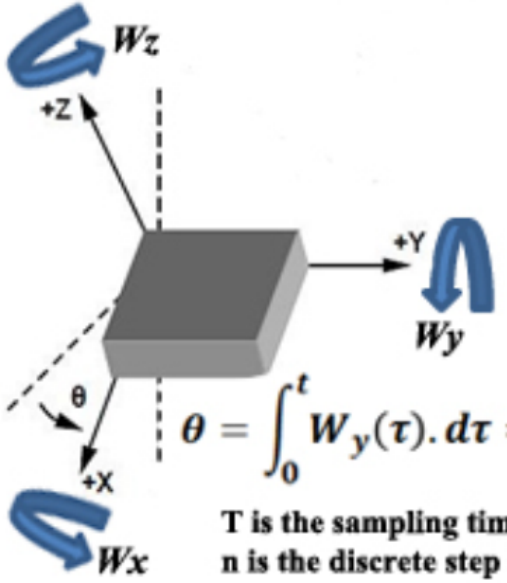
* Orientation Angles:

-> The orientation angle of an object expressed in the sensor body frame can be calculated by integrating the angular velocity measured by the gyroscope over a period of time.

-> In order to get the orientation angles expressed in Euler convention, the following equations can be used:

$$\begin{bmatrix} \frac{\delta pitch}{\delta t} \\ \frac{\delta roll}{\delta t} \\ \frac{\delta heading}{\delta t} \end{bmatrix} = \frac{1}{\cos(pitch)} \cdot \begin{bmatrix} 0 & \cos(roll)\cos(pitch) & -\sin(roll)\cos(pitch) \\ \cos(pitch) & \sin(roll)\sin(pitch) & \cos(roll)\sin(pitch) \\ 0 & \sin(roll) & \cos(roll) \end{bmatrix} \cdot \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix}$$

$$\begin{bmatrix} pitch(t) \\ roll(t) \\ heading(t) \end{bmatrix} = \begin{bmatrix} pitch(t-1) \\ roll(t-1) \\ heading(t-1) \end{bmatrix} + \begin{bmatrix} \frac{\delta pitch}{\delta t} \\ \frac{\delta roll}{\delta t} \\ \frac{\delta heading}{\delta t} \end{bmatrix} . \mathrm{T}$$



$$\theta = \int_0^t W_y(\tau).d\tau = T.\sum_{k=0}^{n} W_y(k)$$

T is the sampling time period in seconds

n is the discrete step at time t

2.3 Temperature sensor characteristics

Table 5. Temperature sensor characteristics

Symbol	Parameter	Test condition	Min.	Typ.(1)	Max.	Unit
TODR	Temperature refresh rate	Gyro OFF(2)		50		Hz
		Gyro ON		59.5		
TSen	Temperature sensitivity(3)			16		LSB/°C
Top	Operating temperature range		-40		+85	°C

1. Typical specifications are not guaranteed.
2. When the accelerometer ODR is set to 10 Hz and the gyroscope part is turned off, the TODR value is 10 Hz.
3. The output of the temperature sensor is 0 (typ.) at 25 °C

Read the raw value from temperature sensor and divide by 16 and then add offset.

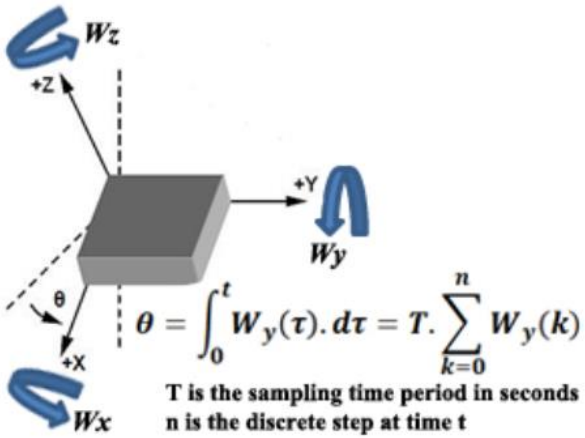
Table 3. Sensor characteristics

Symbol	Parameter	Test conditions	Min.	Typ.(1)	Max.	Unit
LA_So	Linear acceleration sensitivity	Linear acceleration FS = ±2 *g*		0.061		mg/LSB
		Linear acceleration FS = ±4 *g*		0.122		
		Linear acceleration FS = ±8 *g*		0.244		
		Linear acceleration FS = ±16 *g*		0.732		
M_GN	Magnetic sensitivity	Magnetic FS = ±4 gauss		0.14		mgauss/ LSB
		Magnetic FS = ±8 gauss		0.29		
		Magnetic FS = ±12 gauss		0.43		
		Magnetic FS = ±16 gauss		0.58		
G_So	Angular rate sensitivity	Angular rate FS = ±245 dps		8.75		mdps/ LSB
		Angular rate FS = ±500 dps		17.50		
		Angular rate FS = ±2000 dps		70		

Read the raw value from gyro output sensor and mutliply by 70 mili degree per second - offset

angular velocity in deg/sec → gx , gy , gz

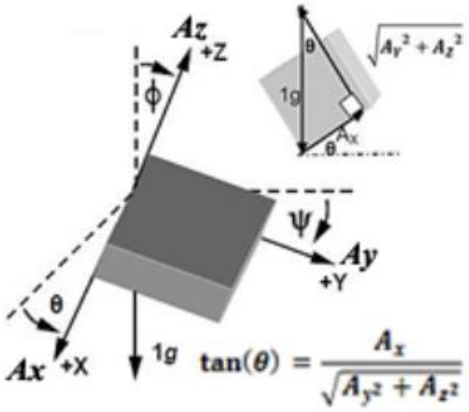
angular position in deg →



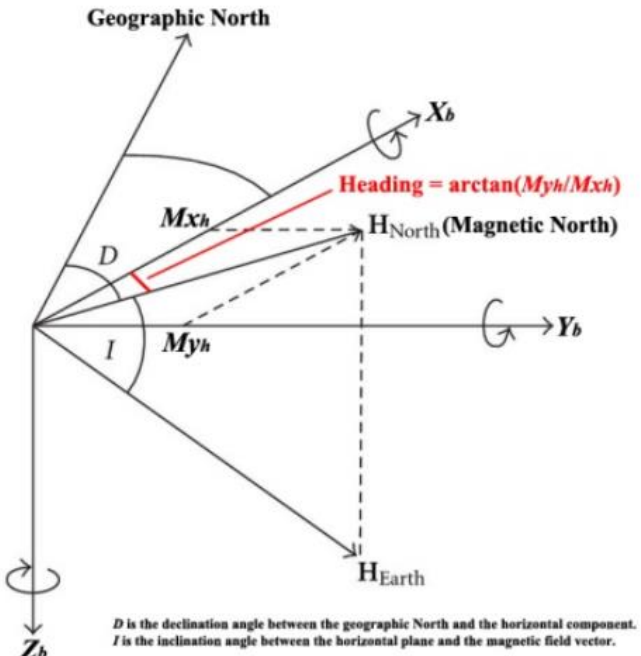
linear acceleration in g → ax , ay , az

magnetic flux density in Gauss → mx , my , mz

the orientation angles of the accelerometer

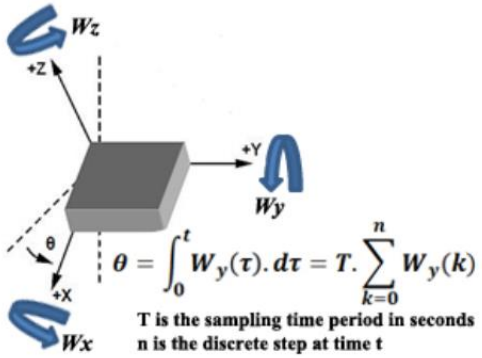


the orientation angles of the magnetometer



orientation angles of the Gyroscope expressed in Euler convention

$$\begin{bmatrix} \frac{\delta pitch}{\delta t} \\ \frac{\delta roll}{\delta t} \\ \frac{\delta heading}{\delta t} \end{bmatrix} = \frac{1}{\cos(pitch)} \cdot \begin{bmatrix} 0 & \cos(roll)\cos(pitch) & -\sin(roll)\cos(pitch) \\ \cos(pitch) & \sin(roll)\sin(pitch) & \cos(roll)\sin(pitch) \\ 0 & \sin(roll) & \cos(roll) \end{bmatrix} \cdot \begin{bmatrix} W_x \\ W_y \\ W_z \end{bmatrix}$$

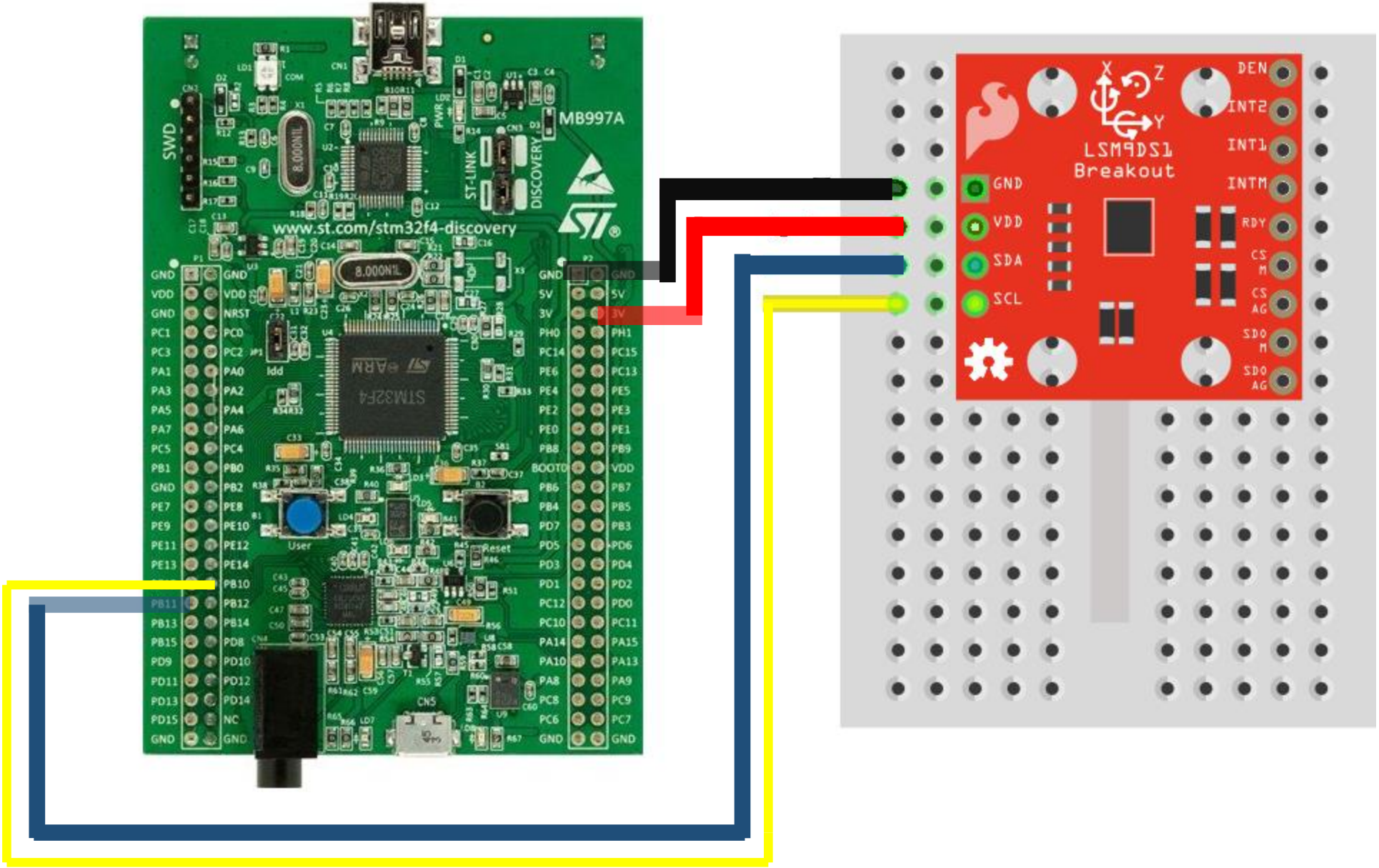


$$\begin{bmatrix} pitch(t) \\ roll(t) \\ heading(t) \end{bmatrix} = \begin{bmatrix} pitch(t-1) \\ roll(t-1) \\ heading(t-1) \end{bmatrix} + \begin{bmatrix} \frac{\delta pitch}{\delta t} \\ \frac{\delta roll}{\delta t} \\ \frac{\delta heading}{\delta t} \end{bmatrix} . \mathrm{T}$$

Exercise:

Inertial Measurement Unit „I2C“

Hardware Setup:



Exercise:

Inertial Measurement Unit „I2C“

UM1472
User manual

Discovery kit with STM32F407VG MCU

Introduction

The STM32F4DISCOVERY Discovery kit allows users to easily develop applications with the STM32F407VG high-performance microcontroller with the Arm® Cortex®-M4 32-bit core. It includes everything required either for beginners or experienced users to get started quickly.

Based on STM32F407VG, it includes an ST-LINK/V2-A embedded debug tool, one ST-MEMS digital accelerometer, one digital microphone, one audio DAC with integrated class D speaker driver, LEDs, push-buttons and a USB OTG Micro-AB connector. Specialized add-on boards can be connected by means of the extension header connectors. The STM32F4DISCOVERY Discovery kit comes with the STM32 comprehensive free software libraries and examples available with the STM32CubeF4 MCU Package.

Figure 1. STM32F4DISCOVERY

Picture is not contractual.

24/32

Table 7. STM32 pin description versus board functions (continued)

STM32 pin			Board function													
Main function	Alternate functions	LQFP 100	Audio DAC	Audio sensor	Motion sensor	Push button	LED	SWD	USB	OSC	Free I/O	Power supply	CN5	CN2	P1	P2
PB9	SPI2_NSS/ I2S2_WS/ TIM4_CH4/ TIM11_CH1/ OTG_FS_SDA/ SDIO_D5/ DCMI_D7/ I2C1_SDA/ CAN1_TX	96	SDA	-	-	-	-	-	-	-	-	-	-	-	-	20
PB10	SPI2_SCK/ I2S2_CK/ I2C2_SCL/ USART3_TX/ OTG_HS_ULPI_D3/ ETH_MII_RX_ER/ OTG_HS_SCL/ TIM2_CH3	47	-	CLK	-	-	-	-	-	-	-	-	-	-	34	-
PB11	I2C2_SDA/ USART3_RX/ OTG_HS_ULPI_D4/ ETH_RMII_TX_EN/ ETH_MII_TX_EN/ OTG_HS_SDA/ TIM2_CH4	48	-	-	-	-	-	-	-	-	PB11	-	-	-	35	-

Exercise:

Inertial Measurement Unit „I2C“

RODOS HAL (I2C):

```
class HW_HAL_I2C; /* hal_i2c.h */

class HAL_I2C : public GenericIOInterface {
HW_HAL_I2C *context;
public:
HAL_I2C(I2C_IDX idx);
 /** Initialize I2C interface default: master, 7Bit address * @param speed I2C bus speed * @retval int32_t zero on success **/
int32_t init(uint32_t speed = 100000);
void reset();
bool isWriteFinished();
bool isReadFinished();

 /** Transmit data: @param addr I2C address of chip, txBuf pointer to transmit buffer, txBufSize size of transmit buffer
  * @retval int32_t number of transmitted bytes **/
int32_t write(const uint8_t addr, const uint8_t *txBuf, uint32_t txBufSize);

 /** Receive data: @param addr I2C address of chip, rxBuf pointer to receive buffer, rxBufSize size of receive buffer
  * @retval int32_t number of received bytes **/
int32_t read(const uint8_t addr, uint8_t *rxBuf, uint32_t rxBufSize);

 /** Send and receive data without a STOP condition between: @param addr I2C address of chip, txBuf pointer to transmit buffer,
txBufSize size of transmit buffer, rxBuf pointer to receive buffer, rxBufSize size of receive buffer
  * @retval int32_t number of received bytes **/
int32_t writeRead(const uint8_t addr, const uint8_t *txBuf, uint32_t txBufSize, uint8_t *rxBuf, uint32_t rxBufSize);
};
```

Example

```
#define LSM9DS1_M_ADDR  0x1E
#define LSM9DS1_AG_ADDR  0x6B

HAL I2C IMU(I2C_IDX2);

IMU.init(400000);


uint8_t LSM9DS1_CTRL_REG1_G[2]= {0x10,0b10011000};

IMU.write(LSM9DS1_AG_ADDR, LSM9DS1_CTRL_REG1_G, 2);


uint8_t  LSM9DS1_WHO_AM_I_G[1] = {0x0F};   /* Gyro WHO_AM_I register address*/

uint8_t DATA[1];

IMU.writeRead(LSM9DS1_AG_ADDR, LSM9DS1_WHO_AM_I_G, 1, DATA, 1);

PRINTF("LSM9DS0_WHO_AM_I_G = %x \r\n", DATA[0]);
```

Exercise:

Inertial Measurement Unit „I2C“

• Tasks:

IMU:

Implement a project with following requirements:

• Implement an initialization function to initialize the sensors to work in specific operation modes by writing to specific registers in the sensors through the I2C interface as following (please refer to the datasheet of the sensors for the corresponding values):

Registers	Addresses	Settings
CTRL_REG1_G	0x10	ODR=238Hz, Cutoff=14Hz , Full Scale =2000
CTRL_REG6_XL	0x20	Acceleration ODR=238Hz, Acceleration Full-scale = +-2g, BW w.r.t ODR, Acceleration Anti-alias Filter Bandwidth=50Hz
CTRL_REG1_M	0x20	ODR = 80 Hz
CTRL_REG2_M	0x21	Magnetic Full-scale = +-4gauss
CTRL_REG3_M	0x22	Continuous Conversion Mode

• Implement three different calibration modes for the three sensors of the IMU that can be activated with telecommands when needed. At the end of each calibration mode, the calculated offset values should be send through UART3.

Exercise:

Inertial Measurement Unit „I2C“

• Tasks:

• Create 3 threads in the project, each of them should toggle different LED when it is running and perform the following:

>> The first thread is the Signal Processing thread that needs to run with following requirements:

- An initialization of the sensors mounted in the extension board should be performed.
- A data gathering process to get the updated values of the three axis of the sensors, should be performed continuously at 10 millisecond time interval using the starting data register addresses of the sensors.

• A signal processing for the gathered information to get the angular velocity in deg/sec, angular position in deg, the linear acceleration in g, and the magnetic flux density in Gauss for all axis and the Temperature in °C using the sensors sensitivity values. In addition, the orientation angles of the Gyroscope expressed in Euler convention and the orientation angles of the accelerometer & magnetometer should be calculated.

Exercise:

Inertial Measurement Unit „I2C“

>>The second thread is the Telemetry thread where the angular velocity in deg/sec, angular position in deg, the linear acceleration in g, the magnetic flux density in Gauss for all axis and the Temperature in °C plus the orientation angles expressed in Euler convention of the Gyroscope and the orientation angles of the accelerometer & magnetometer all in the range between 0-to-360 degree are the telemetry data need to be sent through UART3 when needed.

>>The third thread is the Telecommand thread where the signal processing thread time interval and the telemetry thread time interval should be able to update with telecommands.

Tips:

The **7-bit addresses** for the sensors are: **LSM9DS1_AG (Accelerometer and Gyroscope)=0x6B** and , **LSM9DS1_M(Magnetometer)=0x1E**.

- The **data** of the **IMU** is updated inside **6 registers** in each sensor. In order to retrieve these information, **6 bytes** need to be **read** which represents the **three axis** information as two's complement data in **16-bit format, left justified [OUT_X_L, OUT_X_H, OUT_Y_L, OUT_Y_H, OUT_Z_L, OUT_Z_H]**.

- The **data** of the **temperature sensor** is updated inside **2 registers**. In order to retrieve these information, **2 bytes** need to be **read** which represents the **temperature** information as two's complement data in **12-bit format, right justified [OUT_TEMP_L_XM, OUT_TEMP_H_XM]**.

- The **starting data register addresses** for the sensors can be easily find out from the datasheet of **LSM9DS1**

Tips:

- In order to **write/read multiple bytes** to/from the sensor (I2C Burst Mode), the **MSB (Most Significant Bit)** of the **starting data register address** should be set to **1** to indicate **auto address increment** for the sensor.

In RODOS, HAL_I2C interface's write and writeRead functions do it automatically, you just need to provide The device and register addresses as well as the data to be written in the register.

- In order to get **orientation angles** expressed in **Euler convention** using gyroscope, the **angular velocity in rad/sec** for the three axis need to be calculated using the **sensor sensitivity value** from the **datasheet** that correspond to the used operation scale.

- The **temperature sensor output change vs. temperature** is **8 LSB/°C**.

Calibration Process of Accelerometer

Read the raw accelerometer output at three different orientations

P1 $x = 0, y = 0, z = 1$ orientation

P2 $x = 1, y = 0, z = 0$ orientation

P3 $x = 0, y = 1, z = 0$ orientation

Take 100 samples on all locatons

Calculate offset for each axis

$$\text{offsetX} = \frac{\sum P_{1x} + \sum P_{3x}}{200}; \quad \text{offsetY} = \frac{\sum P_{1y} + \sum P_{2y}}{200} \quad \text{offsetZ} = \frac{\sum P_{3z} + \sum P_{2z}}{200}$$